

SOLOMON SMITH

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EXPERIENCE

Software Engineering Intern

December 2025 to March 2026

Recura AI (courtrules.app), Remote

- Built distributed scraping pipeline ingesting court rules across 50+ U.S. jurisdictions; implemented retry, backoff, and deduplication layers handling 200K+ documents at 99%+ extraction accuracy.
- Shipped retrieval-augmented generation (RAG) system over normalized legal corpus using vector embeddings and hybrid keyword search; reduced query latency from 4s to under 600ms.
- Designed evaluation harness scoring legal-text answer accuracy against human-labeled ground truth; caught a 12% hallucination regression after a model swap and blocked the release.
- Authored Python data pipelines (Postgres, S3, scheduled jobs) and CI checks for schema drift; reduced production data incidents by 70%.

Full-Stack Developer

June 2025 to September 2025

RideSplits, San Bernardino, CA

- Migrated 100% of authentication flows from JWT to Firebase Auth across multiple user roles; reduced login-related bugs by 60%.
- Designed secure multi-screen document upload pipeline with validation and size limits, enabling ID verification for 50+ users.
- Authored security documentation defining Firestore rules and Storage access controls to standardize long-term system governance.

IT Assistant

June 2023 to January 2024

Cañada College, Redwood City, CA

- Improved system reliability by 20% through troubleshooting hardware, software, and networking for 100+ institutional users across Linux, Windows, and macOS lab systems.
- Resolved 90% of technical support requests within 24 hours through remote and in-person assistance.

PROJECTS

SOC Triage AI | Python, Claude Sonnet 4.5, sentence-transformers, Streamlit, pytest

- Built RAG-grounded security alert triage system mapping alerts to MITRE ATT&CK techniques with strict JSON schema validation; achieved 100% pass rate across a 7-case reliability harness.
- Engineered guardrail logic that refuses to triage low-similarity alerts and validates LLM output structurally, preventing fabrication and confident-wrong-answer failures in security contexts.
- Improved harness pass rate from 43% to 100% by diagnosing the root cause as corpus chunking rather than prompt design; documented full reflection in a model card. Source: github.com/SolomonSmith-dev/soc-triage-ai

Multi-Agent AI System (Sauron Stack) | Python, Node.js, Redis, PM2, systemd, Anthropic Claude API, Tailscale, Docker

- Built production multi-agent system on Debian server with PM2-managed services: router (Earendil), executor (Sauron), orchestrator (Morgoth), security daemon (Balrog); 24/7 uptime over Tailscale mesh.
- Diagnosed 401 crash-loop cascade in executor by tracing rate-limit headers; implemented exponential backoff and request queuing to restore stability.
- Resolved better-sqlite3 ABI mismatch by rebuilding under Node v24, unblocking Redis-backed memory store across distributed agents.

Adversarial Search and Intelligent Agents | Python, Pygame

- Implemented Minimax with alpha-beta pruning for a 5x5 Tic-Tac-Toe agent at depth 4 with sub-500ms latency across 1,000+ game states; center-and-corner heuristic cut effective search space by ~40%.
- Solved Constraint Satisfaction Problems for 5 vehicles and 20+ constraints using backtracking with arc-consistency, achieving 100% conflict-free assignment.

SKILLS

Languages: Python, C++, JavaScript, TypeScript, Node.js, SQL, Bash, R, LaTeX

AI / ML: LangChain, Anthropic Claude API, RAG, vector search (Milvus, pgvector), sentence-transformers, scikit-learn, pandas, NumPy, evaluation harnesses, prompt engineering, multi-agent orchestration, MCP

Backend and Data: FastAPI, Express, React, Redis, PostgreSQL, Firebase, Supabase, REST APIs, web scraping at scale, ETL pipelines, schema validation, pytest

Infra and Cloud: Docker, Linux/Debian, systemd, PM2, Tailscale, Nginx, CI/CD (GitHub Actions), Google Cloud Platform, AWS (S3, EC2)

Systems and Tools: Pthreads, IPC, signals, mutex/semaphore, distributed systems, Git, GitHub, Jupyter, Postman, Streamlit, Markdown

EDUCATION

B.S. Computer Science, California State University, San Bernardino

Expected December 2026

GPA 3.16 • Dean's List Spring and Fall 2025 • Coursework: ML, AI, Algorithms, OS, Computer Architecture, Statistics